

PHYS 580 Homework 6 (Chap.8,9)

All questions are worth 10 points, irrespective of complexity or length. Please submit all your solutions on Brightspace in PDF format, together with the source code (if any). Make sure to

- discuss the physics and your results, as appropriate and/or directed,
- include graphical output to illustrate the results,
- include the source codes of your programs (at least the critical parts thereof),
- describe briefly what your program does, and how it does it (algorithm), and
- state the nature of the numerical approximation you used and demonstrate how you know that the approximation (with the parameters you used) is adequate for the problem you used it for.

Note the errata pages for the textbook: <http://www.physics.purdue.edu/~hisao/book/www/errata.pdf>
which are explained further on <http://www.physics.purdue.edu/~hisao/book/www/errata.html>

1. Problem 8.3 (p.257)

Calculate M for the Ising model on a square lattice and try to estimate β . You should find a value close to $1/8$. Repeat this calculation for a triangular lattice. It turns out that β is the same for all regular two-dimensional lattices. However, its value does depend on the dimensionality. (*See hint to this problem in the book.*)

It is enough if you calculate **either** for the square grid, or the triangular one (the latter takes a little more thought). Doing both cases is optional.

2. Problem 8.7 (p.258)

Obtain the specific heat as a function of temperature for a 10×10 square lattice by differentiating the energy and through the fluctuation-dissipation theorem. Show that the two methods give the same result. Which approach is more accurate (for a given amount of computer time)?

3. Problem 8.15 (p.267)

Scaling behavior is found for thermodynamic quantities other than the magnetization. Calculate the susceptibility χ at various values of T and H around the critical point of the Ising model on a square lattice, and study data collapsing using your results. The scaling

form for χ is $\chi(t, h) = |t|^{-\gamma} g_{\pm} \left(\frac{h}{|t|^{\beta\delta}} \right)$, where the critical exponent $\gamma = 7/4$.

4. Problem 9.1 (p.283)

Calculate the speed distributions for a dilute gas as in Figure 9.4 and compare the results quantitatively with the Maxwell distribution. (For example, perform the χ^2 analysis described in Appendix G.) This analysis also yields the temperature; compare the value you find with the result calculated directly from the equipartition theorem, $k_B T = \left\langle \frac{m}{2} (v_x^2 + v_y^2) \right\rangle$.